

North American Actuarial Journal

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/uaaj20

Enhancing Mortality Forecasting through Bivariate Model–Based Ensemble

Liqun Diao, Yechao Meng, Chengguo Weng & Tony Wirjanto

To cite this article: Liqun Diao, Yechao Meng, Chengguo Weng & Tony Wirjanto (09 Mar 2023): Enhancing Mortality Forecasting through Bivariate Model–Based Ensemble, North American Actuarial Journal, DOI: 10.1080/10920277.2023.2167832

To link to this article: https://doi.org/10.1080/10920277.2023.2167832

Published online: 09 Mar 2023.

Submit your article to this journal

Article views: 119

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=uaaj20

North American Actuarial Journal, 0(0), 1–20, 2023
© 2023 Society of Actuaries
ISSN: 1092-0277 print / 2325-0453 online
DOI: 10.1080/10920277.2023.2167832

Routledge
Taylor & Francis Group

Check for updates

Enhancing Mortality Forecasting through Bivariate Model–Based Ensemble

Liqun Diao, Yechao Meng,* Chengguo Weng, and Tony Wirjanto
Department of Statistics and Actuarial Science, University of Waterloo, Waterloo, Ontario, Canada

We propose a bivariate model–based ensemble (BMBE) method to borrow information from the mortality data of a given pool of auxiliary populations to enhance the mortality forecasting of a target population. The BMBE method establishes a cascade of bivariate mortality models between the target population and each auxiliary population as the base learners. Then it aggregates prediction results from all of the base learners by means of an averaging strategy. Augmented common factor–type and CBD-type bivariate models are applied as the base learners as illustrative examples in the empirical studies with the Human Mortality Database. Empirical results presented in this article confirm the effectiveness of the proposed BMBE method in enhancing mortality prediction. For completeness, we also conduct a synthetic study to illustrate a particular setting for the superior performance of the BMBE method.

1. INTRODUCTION

Longevity risk, attributed to the increase in human life expectancy, has long been recognized as one of the major risks in nature faced by insurers, governments, and individuals alike. The effect of longevity risk is systematic, and it has created substantial financial pressure on the pension fund industry and annuity providers. Mortality modeling is critical to managing longevity risk, and the past decades have witnessed a steady proliferation of various mortality models proposed in the literature. The Lee-Carter model (Lee and Carter 1992) has notably become the benchmark model for single-population mortality modeling and prediction.

As a prototype of multi-population mortality models, the augmented common factor (ACF) model proposed by N. Li and Lee (2005) extends the Lee-Carter model to multiple populations. Further development of multipopulation mortality models includes J. S.-H. Li and Hardy (2011), Russolillo, Giordano, and Haberman (2011), J. Li (2013), Wang and Yang (2013), Zhou et al. (2014), and Kleinow (2015), among others. One strand of literature on multi-population mortality modeling is based on an age–period–cohort (APC) structure of mortality rates, which is a time-honored methodology in epidemiology. The representative literature from this strand includes Dowd et al. (2011) and Janssen, Wissen, and Kunst (2013). The idea of the APC structure–based models is to describe a mathematical relationship between death rates and attained age and that between calendar period and birth cohort. A functional data approach has also been adopted for multipopulation mortality modeling (e.g., Hyndman, Booth, and Yasmeen 2013; Shang and Hyndman 2017). Moreover, J. S.-H. Li, Zhou, and Hardy (2015) proposed bivariate-population generalizations for single-population models, denoted respectively as M1–M3 and M5–M8, from Cairns et al. (2009).

Some studies have documented that multipopulation mortality forecasting models are generally superior to individual mortality forecasting models when the populations included in the joint model share similar patterns in their trajectories of mortality development (e.g., N. Li and Lee 2005; Shang, 2016; J. S.-H. Li, Chan, and Zhou 2017; Shair, Purcal, and Parr 2017). The similarity in socioeconomic conditions and geographic proximity have been proposed as criteria for the choice of populations to be modeled jointly, but these criteria in general are vague and require subjective judgment in applications, as pointed out by N. Li and Lee (2005).

In this article, we propose an ensemble statistical learning method that allows for borrowing information from the mortality data of a given pool of auxiliary populations to enhance the accuracy of the mortality forecast for a target population. In this

Address correspondence to Chengguo Weng, Department of Statistics and Actuarial Science, University of Waterloo, M3-200 University Avenue West, Waterloo, ON N2L 3G1, Canada. E-mail: c2weng@uwaterloo.ca

*Additional Affiliation: School of Mathematical and Computational Sciences, University of Prince Edward Island, Charlottetown, Prince Edward Island, Canada.

ensemble method, we use bivariate mortality models as the base learners. We establish a sequence of bivariate mortality models between the target population and each auxiliary population from the given pool and then apply an "averaging" strategy to group all of the estimates from those established bivariate models.

The idea of averaging over a sequence of estimates for the same prediction target was commonly used in meta-analysis in statistics and stacking ensemble methods in machine learning (e.g., Raftery, Madigan, and Hoeting 1997; Claeskens and Hjort 2008; Hansen and Racine 2012; and references therein). For mortality prediction, model averaging methods have been adopted by Shang (2012), Shang and Haberman (2018), and Shang and Booth (2020), and ensemble methods have been used in Kessy et al. (2021) and Chang and Shi (2022). These studies fit different types of single-population mortality models to the mortality data of a given population and then apply certain model averaging strategies to aggregate the forecasts from various models to obtain a final forecast. In contrast, we utilize the averaging approach as a novel means to borrow information from multiple auxiliary populations to enhance the forecasting accuracy for a target population. We aggregate predictions for a target population that are obtained from a cascade of bivariate-population models of the same type (i.e., base learners), each built upon the data of the target population and one of the auxiliary populations. The bivariate model–based ensemble (BMBE) method, unlike the standard setting of model averaging in the aforementioned literature, plays a similar role as the subsample aggregation method, also known as "subagging," proposed by Bühlmann and Yu (2002). The main feature of the method lies in aggregating the predictive results from the same type of base learner on different subsamples.

Our proposed BMBE method facilitates borrowing information from multiple populations with several distinct advantages over the traditional multipopulation modelling. First, the BMBE method helps reduce the risk of model misspecification. In real applications, model misspecification may inevitably exist. In particular, a higher dimensional multipopulation model tends to have a greater amount of risk of misspecification than a bivariate model unless we know the true underlying structure of a joint model precisely. Second, the BMBE method circumvents the potential computational difficulty and intractability of calibrating a large-dimensional multipopulation model. In our study we add a time shift parameter to the ACF and Cairns-Blake-Dowd (CBD) models to reflect the delay or advance of mortality development by one population over another. Calibrating a high-dimensional multipopulation model of this sort is computationally challenging and also vulnerable to the issue of instability in numerical computation. However, it is computationally tractable for a bivariate such model with a time shift parameter. Third, the BMBE method has the inherent flexibility of being able to work with different base learners. Particularly, when bivariate-population models such as ACF and CBD models are adopted as the base learner, the flexibility of the BMBE method allows us to add the time shift parameter into the common mortality trend component to characterize the time by which one population is ahead of or behind the other in terms of their mortality development stages.

According to Zhou et al. (2014), some populations tend to be more dominating, whereas others tend to follow their mortality dynamics. For instance, a developing country may demonstrate a similar mortality improvement pattern in the most recent decade to what a developed country experienced in an earlier decade. Furthermore, as mentioned in section 5.4 of J. S.-H. Li, Zhou, and Hardy (2015), for two populations, the period effect estimates, which capture the mortality developing trend in time, usually roughly co-move with one another but are located at different absolute levels. Leads and lags in mortality have also been studied in Milidonis and Efthymiou (2017) with the view that populations in developed countries would lead the mortality changes in populations in less developed countries. These observations motivate us to study the effect of time shift on mortality forecast under our proposed BMBE framework.

In our study, we propose a data-driven "rank and average" strategy that ranks auxiliary populations according to their ability to improve the accuracy of the mortality forecast for the target population and averages over those top-performing ones selected by a conventional validation procedure. We also investigate several other "averaging strategies" for comparison purposes, including a simple average over the whole pool of auxiliary populations, an average within geographical subgroups, an average within k-means clusters, and an average targeting to reach the minimum sum of squared errors.

In this article, we apply the proposed BMBE prediction method to mortality data of 24 populations of both genders from the Human Mortality Database (HMD). We conduct two empirical studies. In the first one, we adopt the ACF model (with and without a time shift component) as the base learner and apply the BMBE method with various averaging strategies to predict mortality rates of ages between 0 and 100. In the second one, we take the CBD model (with and without a time shift component) as the base learner because our goal here is to predict mortality rates for seniors aged 55 to 90. The outperformance of the BMBE method over several benchmark methods is confirmed in both empirical studies, particularly when the time shift parameter is included in the base learners. Furthermore, we analyze the effect of the inclusion of a cohort effect term in the base learners in both empirical studies. In addition, we conduct a synthetic study to illustrate a scenario where high-dimensional multipopulation models tend to suffer from greater model misspecification than bivariate models for each pair of neighboring populations.

The empirical studies in the article reveal several important points of using the BMBE method. First, if the utilized base learner is a misspecified bivariate-population model for most pairs of populations, the biases arising in individual bivariate-population

models accumulate, and this can lead to poor prediction results in the BMBE method. Lessening the extent of misspecification in base learners is a key to attain better performance when using the BMBE method. Second, a weak base learner with fewer model assumptions tends to predict less accurately than a strong base learner with more model assumptions provided that these model assumptions are correctly specified. However, the weak base learner performs better when these model assumptions are not satisfied, so there is a distinct trade-off between prediction accuracy and robustness. Third, the rank and average strategy turns out to be more robust to the misspecification of the base learner compared to other averaging strategies.

The rest of the article proceeds as follows. Section 2 describes the ACF and CBD models with the inclusion of a time shift parameter. These two models are used as the base learners in our empirical studies. Section 3 introduces several averaging strategies. Section 4 presents empirical studies with the HMD and a synthetic study. Finally, Section 5 provides concluding remarks and some discussions on possible avenues for future research.

2. THE BASE LEARNER

In our proposed BMBE framework, the base learner is a bivariate-population model used to link the target population with each auxiliary population from a given pool. In principle, any bivariate-population model can be used as the base learner in our proposed BMBE framework. In our study, we use two prevailing bivariate-population models, the ACF and CBD models, as the testbed because of their unique historical roles in the literature. As mentioned in the preceding section, we add a time shift parameter Δt to the bivariate-population models to investigate how it can affect the resulting forecasting accuracy of our BMBE approach. We label the resulting bivariate-population models CF-ts and CBD-ts, with the suffix indicating the inclusion of the time shift parameter in the models.

2.1. ACF-ts Model

Let $\log m_j(x,t)$ denote the central logarithmic mortality rate at age x in year t of the jth population, for $x = x_1, ..., x_n$, $t = 0, 1, 2...$, and $j = 1, 2$. The ACF-ts model is an extension of the bivariate-population ACF model of N. Li and Lee (2005) with an extra parameter Δt representing the time shift:

$$\log m_1(x,t) = a_1(x) + B(x)K(t) + b_1(x)k_1(t) + \epsilon_1(x,t), \tag{1}$$

$$\log m_2(x,t) = a_2(x) + B(x)K(t-\Delta t) + b_2(x)k_2(t) + \epsilon_2(x,t). \tag{2}$$

The first population (with subscript 1) is the target population for mortality prediction. The second population (with subscript 2) is the population included to borrow information from and is hereafter referred to as the auxiliary population. The time shift parameter Δt, which takes integer values from an interval symmetric about 0, characterizes the advance or delay of "evolution" in years by the target population over the auxiliary. If $\Delta t = 0$, the model degenerates to the bivariate-population ACF model of N. Li and Lee (2005). The other components in the model carry the same meaning as those in the ACF model; that is, the common trend term is a product of $B(x)$, a deterministic age function, and $K(t)$, a stochastic period function, $b_j(x)$ and $k_j(t)$ constitute the population-specific components, a_j represents the average of the age-specific logarithmic mortality rates over a certain historical period for population j, and $\epsilon_j(x,t)$ are the white noise terms, $j = 1, 2$. The time lag structure of the common process in the aforementioned base learner is a special case of a more general model presented in Milidonis and Efthymiou (2017) for the Granger causality test.

Assume that we have mortality data of both populations for year $t \in \{0, 1, ..., T\}$ with a positive integer T. To avoid the issue of model unidentifiability, we impose the following constraints in parallel to the ACF model:

$$\sum_{l=1}^{n} B(x_l) = 1, \quad \sum_{t\in\mathbb{S}} K(t) = 0, \quad \sum_{l=1}^{n} b_j(x_l) = 1, \quad \text{and} \quad \sum_{t=0}^{T} k_j(t) = 0,$$

where $\mathbb{S} = \{-\Delta t, -\Delta t + 1, ..., T\}$ for $\Delta t \geq 0$ and $\mathbb{S} = \{0, 1, ..., T - \Delta t\}$ for $\Delta t < 0$. In our study, we fit the common trend sequence $K(t)$ with a random walk process with drift (RWD), which is a widely used model in the literature. Meanwhile, we fit the population-specific components $k_j(t)$ with the "best" autoregressive integrated moving average (ARIMA) model chosen according to the Akaike information criterion. For the calibration of the ACF-ts model, we prespecify a set of integer values for Δt, implement the usual singular value decomposition (SVD) procedure for the calibration of the model with Δt fixed at

each integer from the prespecified set, and then apply a validation procedure to choose the best Δt value (see Section 2.3). We relegate the step-by-step calibration procedure to Appendix A.1.

2.2. CBD-ts Model

Let $q_j(x,t)$ denote the probability that an individual in population j aged x will die between t and $t+1$ given the individual is alive at time t, for $t = 0, 1, ...,\ x = x_1, ..., x_n,$ and $j = 1, 2$. The CBD-ts model describes the mortality development of two populations as follows:

$$\text{logit}(q_1(x,t)) = K(t) + (x - \bar{x})k_1(t) + \epsilon_1(x,t), \tag{3}$$

$$\text{logit}(q_2(x,t)) = K(t - \Delta t) + (x - \bar{x})k_2(t) + \epsilon_2(x,t), \tag{4}$$

where $\text{logit}(q) \equiv \log\left(\frac{q}{1-q}\right)$, $\bar{x} = \sum_{l=1}^{n} x_l/n$. The time shift parameter Δt takes integer values from a symmetric interval centered at 0. $K(t)$ is a stochastic process that captures the common mortality development of both populations. $k_1(t)$ and $k_2(t)$ are two stochastic processes reflecting the population-specific development of mortality. This specification extends the original single-population CBD model of Cairns, Blake, and Dowd (2006) or, equivalently, model M5 in Cairns et al. (2009) to a bivariate-population model by introducing a common age–period term $K(t)$ shared by both populations and the time shift parameter to reflect the differences in their mortality development stages. We fit the common trend sequence $K(t)$ with an RWD model and the population-specific components $k_j(t)$ with a bivariate RWD model. The calibration of Δt is also carried out through a validation procedure as in the ACF-ts model (see Section 2.3 for details), and that of other components uses the maximum likelihood method (see Appendix A.2 for details).

2.3. Choice of Δt

As mentioned earlier, the parameter Δt is designed to capture the difference in years of mortality development between the target population and the auxiliary population in the bivariate ACF-ts or CBD-ts model. The way to determine the value of Δt is in the same spirit as choosing values of hyperparameters in statistical learning, which can be realized by applying a validation procedure. As part of preparation, the training dataset is partitioned into a modeling set and a validation set and a grid of integer values is prespecified for Δt. The modeling set is used to calibrate a model with a fixed Δt, and this modeling procedure yields a series of fitted models corresponding to each integer in the prespecified set. The value of Δt is determined as the one that yields the smallest overall sum of squared errors (SSE) calculated based on the validation set for predicting $\log m_j(x,t)$ in the ACF-ts model or $\text{logit}(q_j(x,t))$ in the CBD-ts model.

In the analysis of HMD in Section 4, we take a prespecified set of integers over $[-10, 10]$ for Δt. Given two specific populations, sometimes it is clear that the mortality development stage of the target population is in advance of the auxiliary population or the other way around, so a set of positive or negative integers seems appropriate. However, we uniformly adopt a candidate set $\{-10, -9, ..., 9, 10\}$ for Δt when jointly modeling any pair of target and auxiliary populations to make the choice of Δt fully driven by data and avoid selection biases arising from subjective judgment on the development stages of populations.

3. BIVARIATE MODEL–BASED ENSEMBLE FOR PREDICTION

The bivariate-population models introduced in Section 2 allow us to borrow information from one auxiliary population for the forecasting of the target. To borrow information from multiple populations, it is natural to think of a multipopulation model, which, however, can be highly intractable and computationally prohibitive when it comes to calibration. Furthermore, identifying a set of auxiliary populations conducive to enhancing forecast accuracy for the target population is an arduous task. So, instead of calibrating a multipopulation model, we propose an ensemble framework by using the bivariate-population models as the base learners.

The main idea underlying the ensemble procedure is to reduce the prediction uncertainty and improve prediction accuracy by aggregating prediction results from multiple predictive models. Therefore, for a given target population and a pool of auxiliary populations, we fit a bivariate-population model between the target population and each auxiliary population in the pool and obtain the extrapolative results (i.e., forecast) on future mortality rates of the target population from each of the resulting bivariate-population models. We then aggregate all of the forecasts to form a final forecast for the mortality of the target

population. Let $\hat{\eta}_1^{(s)}(x,t)$ denote the mortality forecast of the target population for age x and calendar year t using information from the sth auxiliary population, $s = 1, ..., S$. Here, $\hat{\eta}_1(x,t)$ is a forecast of $\eta_1(x,t) = \log m_1(x,t)$ if the ACF-ts model was adopted as the base learner and $\eta_1(x,t) = \log\left[\frac{q_1(x,t)}{1-q_1(x,t)}\right]$ if the CBD-ts model was used. Then the final ensemble prediction takes a general form of

$$\hat{\eta}_1(x,t) = \sum_{s=1}^{S} w_s \hat{\eta}_1^{(s)}(x,t)\,, \tag{5}$$

where w_s is the weight assigned to the sth auxiliary population. The values of w_s should be determined to ensure a good performance of the resulting prediction. In our study, we design the following rank and average strategy and our empirical studies (see Section 4) confirm a superior performance compared with many benchmark models:

- **Rank and Average (RankAvg)**: This strategy is inspired by the notion that "relevant" populations tend to bring in more information, whereas "irrelevant" populations bring in more noise than useful information. We rank auxiliary populations according to the validation SSEs of the corresponding bivariate-population models on the validation dataset mentioned in Section 2.3. This strategy tends to only keep the top u^* bivariate-population models for the final forecast. For any $u \in \{1, 2, \ldots, S\}$, the final forecast based on the top u bivariate-population models is calculated by assigning $w_s = \frac{1}{u}$ for each of the top u bivariate-population models and $w_s = 0$ for the remaining bivariate-population models in Equation (5). The value of u^* is selected as the one such that the final forecast yields the smallest validation SSE.

The RankAvg strategy suitably addresses the challenge of auxiliary populations selection and facilitates a customized selection procedure for each target population. It is expected to have a relatively more robust performance compared with the other strategies (introduced later) because it takes in the whole pool of auxiliary populations and has a mechanism for screening out those that are impotent in improving the prediction accuracy for the mortality of the target population.

In addition, we consider the following averaging strategies as benchmarks for the purpose of comparision:

- **Simple Average (SimAvg)**: This is the most naive strategy where we assign equal weights to all S auxiliary populations; that is, set $w_s = \frac{1}{S}$, for $s = 1, \ldots, S$, in Equation (5). This strategy does not involve any screening over the cascade of bivariate-population models regarding their efficacy in predicting the mortality of the target population.
- **Average Based on Geographic Information (GeoAvg)**: We use a geographic proximity as exogenous information to **preselect** groups. For a given target population, the final forecast is the average of forecasts from the bivariate-population models with auxiliary populations being within the same geographic group as the target population. Specifically, if there are R auxiliary populations within the same geographic group as the target population, we set $w_s = \frac{1}{R}$ in Equation (5) for the R auxiliary populations and $w_s = 0$ for those auxiliary populations outside the geographic group.
- **Average Based on Clustering Results (KmeansAvg)**: This is a data-driven strategy to prespecify groups by using cluster analysis (k-means method) to find populations with similar mortality characteristics (Hatzopoulos and Haberman 2013). For a given target population, we compute the final forecast as the simple average of the forecasts from the bivariate-population models built with each auxiliary population located within the same cluster as the target.
- **Minimum SSE Averaging (MSSEAvg)**: This strategy determines the weights by minimizing the validation SSE:

$$\sum_{x\in\mathcal{X},\, t\in\mathcal{T}_{validate}} (\eta_1(x,t) - \hat{\eta}_1(x,t))^2,$$

subject to the constraints that $0 \le w_s \le 1$ and $\sum_{s=1}^{S} w_s = 1$, where $\mathcal{T}_{validate}$ denotes the set of years over the period of the validation dataset.

Some of the above benchmark averaging strategies do not necessarily perform well by design, and they were included only for the purpose of comparison; for example, the simple average. The GeoAvg strategy is included to show how the method performs when the combining rule is determined by exogenous information (or more precisely the geographic proximity), and the KmeansAvg is intended to show the performance when using clustering for a combining task. When the validation SSE can fully reflect the true SSE, MSSEAvg should yield the best weights for the best test SSE. In real applications, however, there are more or fewer estimation errors and model misspecification. The validation SSE can only reflect the magnitude of

the out-of-sample SSE to some degree (which is hard to quantify because we do not know the true model underlying the mortality data). Because of the existence of estimation error and model misspecification risk, the weights solved from the MSSEAvg strategy do not necessarily guarantee the best test SSE. This is not unlike the fragility of Markowitz's optimal mean–variance portfolio, where the optimal portfolio weight solved from an estimated mean–variance portfolio could perform rather poorly and often inferior to the simple $1/N$ portfolio that allocates the same amount of capital to every individual asset in the portfolio, particularly when the number of assets is large (DeMiguel, Garlappi, and Uppal 2009).

4. EMPIRICAL ANALYSIS

In this section, we evaluate the performance of various averaging strategies and base learners under the proposed BMBE framework by applying them to the HMD and comparing them with several benchmark models. We conduct two empirical studies. The first study focuses on the mortality forecast for a full range of ages between 0 and 100 and adopts the bivariate ACF model (with and without time shift) as the base learner. The second study focuses on mortality forecasting for the senior age group, ages 55 to 90, and applies the bivariate CBD model (with and without the time shift component) as the base learner because the CBD model is known for its superior performance in characterizing mortality development for seniors. We also study the influence of adding a cohort effect to the ACF and CBD base learners in our BMBE method.

The first empirical study confirms a noticeable improvement in mortality forecasting accuracy by the BMBE method over benchmark models and a positive contribution in reducing forecast error by adding the time shift component to the base learner. The second empirical study still shows an improvement by the BMBE method although not as significant as in the first empirical study, and the inclusion of the time shift parameter only has a marginal effect on the performance of the resulting mortality forecast. Furthermore, these empirical studies together reveal the importance of avoiding severe misspecification in the base learner for the proposed ensemble method to work well.

In addition, we conduct a synthetic study to illustrate a numerical setting where our BMBE method with the rank and average strategy works better than the higher dimensional multipopulation model–based method.

4.1. Data Description

To strike a balance among different factors, such as the size of the candidate pool, the length of time, and the number of missing values in HMD, we concentrate on the mortality data of 30 populations from 1970 to 2010 in our empirical studies. The mortality data (1970–2010) are split into a training set (1970–2002) and a test set (2003–2010). The training set is further split into a modeling set (1970–1994) and a validation set (1995–2002). We use the modeling set to fit bivariate-population models and create candidate predictive rules and the validation set to determine the value of the time shift parameter Δt in each bivariate-population model. The validation set is also used to determine the optimal number of top-performing forecasts used in the RankAvg strategy.

According to the account reported in Brainerd and Cutler (2005), demographic disasters in the form of sharply rising death rates occurred in several member countries of the former Soviet Union in the 1990s and beyond. These populations reflect an obviously different trajectory of mortality development from the rest in the HMD; thus, one population from the group cannot be seen as an advance or a delay of a population outside the group, which imposes a major challenge in adopting bivariate-population models with a time shift component. Before we hastily exclude these populations from our studies, we conduct further analysis to ensure the appropriateness of the criterion of population choice used in our studies. These empirical studies together also reveal the importance of the exclusion.

We illustrate the age-aggregated logarithmic mortality sequence for each of the 30 populations in the left panel of Figure 1. We then conduct a k-means cluster analysis to detect structural dissimilarity among these sequences of age-aggregated logarithmic mortality. The k-means cluster procedure is designed to exclude the effect of mortality level on the dissimilarity among different populations. Details of the clustering procedure are as follows:

1. Define a dissimilarity matrix $\mathbf{D}$ with entries $D_{i,j}$ being the variance of the sequence $\text{diff}_{i,j}(t) = \sum_x \log m_i(x,t) - \sum_x \log m_j(x,t)$, $t = 1970, 1971, ..., 2002$.
2. Apply multidimensional scaling (MDS) to transform the matrix $\mathbf{D}$ into a two-dimensional objective. According to Cox and Cox (2008), MDS outputs a configuration of points in a two-dimensional space, where each point represents one original sequence and the distance between each point pair retains the dissimilarity measured by $D_{i,j}$ to the greatest extent.
3. Implement a k-means cluster analysis to the transformed outputs of MDS.

It is worth noting that variance is used in the above procedure as the dissimilarity measure so that the disparity in mortality level does not have any effect on the dissimilarity measure. If two populations i and j have parallel trajectories in their age-aggregated logarithmic mortality, the difference sequence is a constant and thus $D_{i,j} = 0$; otherwise, we anticipate a positive value for the variance.

We display the clustering results in the right panel of Figure 1. From the figure, most member countries (Belarus, Bulgaria, Latvia, Lithuania, Russia, and Ukraine) of the former Soviet Union are classified into a separate group. We view these populations as "outliers" in mortality development patterns. We exclude these six populations from our analysis and focus on the remaining 24 populations in our empirical studies; see Table 1 for the specific names of the 24 populations and their geographic grouping (also see Diao, Meng, and Weng 2021; Richman and Wüthrich 2021).

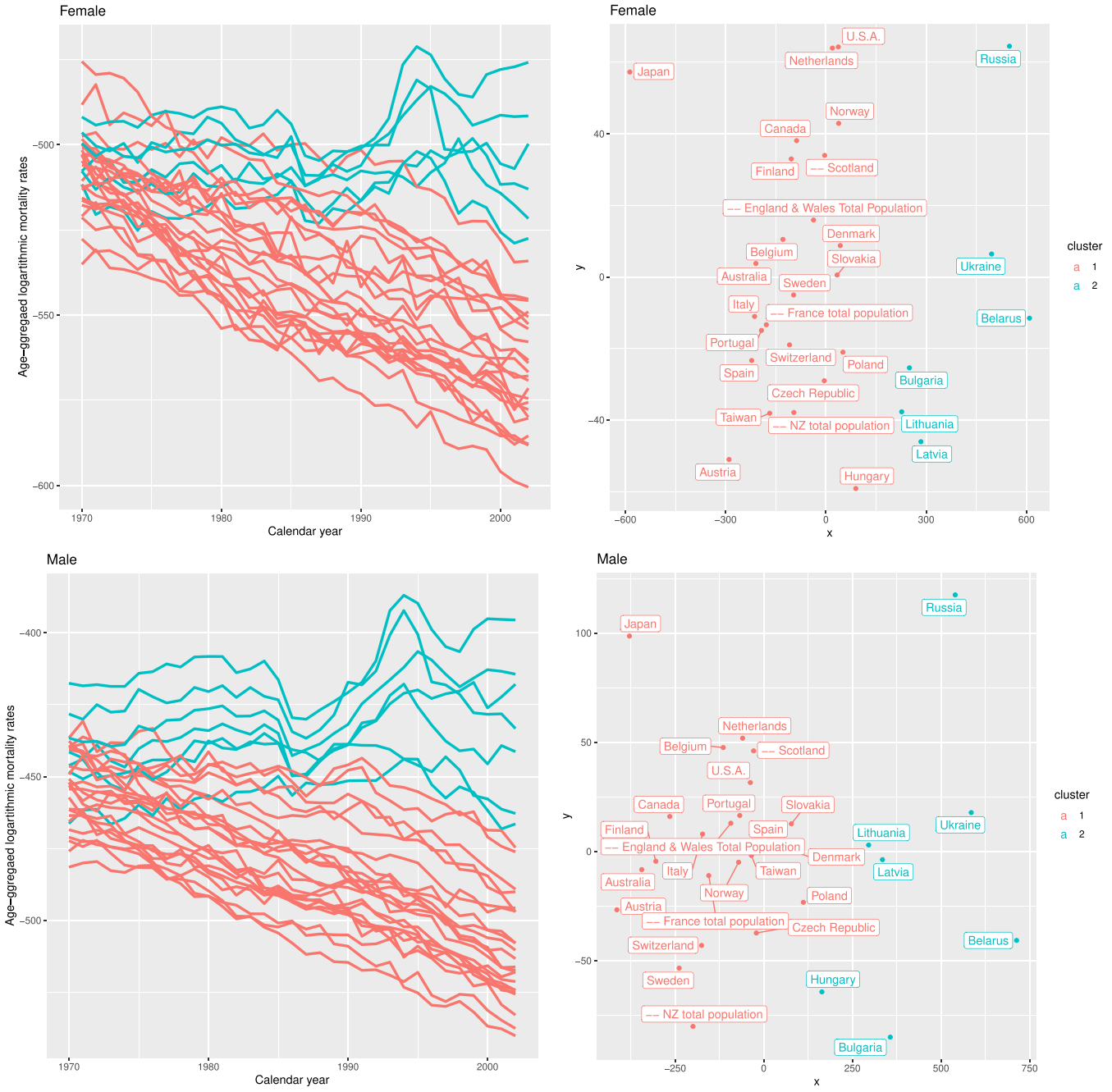


FIGURE 1. Clustering Results of Age-Aggregated Logarithmic Mortality Rates based on MDS. *Note:* From top to bottom: female and male populations.

4.2. Empirical Study with the ACF-ts Model

In this empirical study, we investigate the performance of the BMBE method using the ACF model (with and without the time shift component) as the base learner, study various averaging strategies and assess the value of adding the time shift term, and compare their performance with the classic ACF models in predicting the mortality rates of ages 0 to 100.

4.2.1. Prediction Models

We take each of the 24 populations in Table 1 as the target for mortality forecasting and obtain the forecasts through the BMBE method as described in Section 3. Specifically, we develop a bivariate-population ACF-ts model between the target population and each of the remaining 23 populations, resulting in 23 prediction rules for the mortality of the target population. We then apply an averaging strategy to ensemble all of the 23 prediction results into a final forecast. We apply the five averaging strategies introduced in Section 3 and label the resulting forecasts as ACF-ts·SimAvg, ACF-ts·GeoAvg, ACF-ts·KmeansAvg, ACF-ts·MVWAvg, and ACF-ts·RankAvg, respectively. For ACF-ts·KmeansAvg, we apply a k-means clustering analysis with $k = 8$ and 1000 independent random initializations. We execute the analysis over the 24 populations for both genders.

To examine the value of including the time shift term in the ACF model, we consider the **ACF·RankAvg** model, which is the special case of **ACF-ts·RankAvg** with Δt fixed at zero. We also consider the following benchmark models for comparative analysis:

- **Lee-Carter**: The Lee-Carter model is fitted to each population separately. The sequence of $k(t)$ obtained from an SVD procedure is fitted using the `auto.arima` function from the `R` package `forecast` (Hyndman and Khandakar 2007) to search for a suitable ARIMA model.
- **ACF·AIO**: The ACF model is fitted to the 24 populations jointly.
- **ACF·GeoInfo**: The ACF model is fitted to each geographic groups in Table 1.
- **ACF·kmeans**: The ACF model is fitted to each cluster from the k-means clustering algorithm (with $k = 8$ and 1000 independent random initializations).

We calibrate each of the above benchmark models by using the training set (1970–2002) and then extrapolate the resulting models to the testing period (2003–2010) for mortality forecasting.

4.2.2. Prediction Performance Evaluation Based on Test SSEs

The prediction accuracy is evaluated in terms of test SSEs, which are denoted by $e(t)$ and calculated as follows:

$$e(t) = \sum_{x=0}^{100} [\log m_1(x,t) - \log \hat{m}_1(x,t)]^2, \quad t = 2003, ..., 2010,$$

TABLE 1
Geographic Grouping of 24 Populations from HMD

Target population	Geographic group	Target population	Geographic group
Australia	Oceania	Netherlands	Western Europe
Austria	Western Europe	New Zealand	Oceania
Japan	Asia	Norway	Scandinavia
Belgium	Western Europe	Poland	Eastern Europe
Scotland	Great Britain	Portugal	South Europe
Canada	North America	United States	North America
Czech Republic	Eastern Europe	Slovakia	Eastern Europe
Denmark	Scandinavia	Spain	South Europe
Finland	Scandinavia	Sweden	Scandinavia
France	Western Europe	Switzerland	Western Europe
Hungary	Eastern Europe	Taiwan	Asia
Italy	Southern Europe	England and Wales	Great Britain

where $m_1(x,t)$ is the realized mortality rate of a target population, and $\hat{m}_1(x,t)$ is its forecast. For a succinct summary about the performance of each forecasting model, we compute the overall test SSE as $\sum_{t=2003}^{2010} e(t)$ for each target population. A smaller test SSE indicates a better prediction performance.

Rotating the target population over the 24 in the pool, we obtain 24 overall test SSEs. We report the first quartile, median, mean, and third quartile of the 24 overall test SSEs for each forecasting model in Table 2. These results indicate that most of our proposed BMBE based forecasts (the bottom five in each panel of Table 2) except ACF-ts·MSSEAvg, particularly ACF-ts·RankAvg and ACF-ts · SimAvg, substantially outperform the benchmark models (the top four models in each panel of Table 2) with a smaller median and mean of the resulting test SSEs. The first and third quartiles from each of the five BMBE-based predictions are also generally smaller than those for the benchmark models.

The 24 populations show a similar downward trend in their trajectories of logarithmic mortality rate as shown by the red curves in Figure 1. Therefore, the trajectory of one population may be well approximated by shifting that of another population and the ACF model with a time shift term can be considered a good fit for most pairs of populations. When the base learner is a correctly specified model, the SimAvg strategy is expected to perform well because it has the advantage of averaging over the full list of auxiliary populations and makes use of all useful information. The relative underperformance of the GeoAvg and KmeansAvg strategies is due to the fact that they both only borrow information from a small subset of the population pool. The RankAvg strategy gives comparable results to the SimAve strategy. A comparison between ACF-ts·RankAvg and ACF·RankAvg confirms the benefit of including a time shift term in the ACF base learner.

Evaluation Based on One-Sided Diebold-Mariano Tests We conduct a one-sided Diebold-Mariano (DM) test to determine whether the test SSE from one predictive model is statistically significantly smaller than that of another. The DM test is a commonly used statistical hypothesis testing method in forecast comparison; see Diebold and Mariano (2002) and Harvey, Leybourne, and Newbold (1997).

For a pair of models in comparison, Model A and Model B, we apply two one-sided DM tests, with the null hypothesis that Model A is no worse than Model B and that Model B is no worse than Model A. The hypothesis tests are conducted based on

TABLE 2
Summary Statistics of Test SSEs Comparing the Prediction Performance of the BMBE-Based Approaches Using ACF or ACF-ts as the Base Learner versus Benchmark Models

	First quartile	Median	Mean	Third quartile
		Female population		
Lee-Carter	19.24	45.11	44.04	63.86
ACF·AIO	20.49	44.91	43.76	62.09
ACF·GeoInfo	20.95	45.49	42.96	63.77
ACF·kmeans	18.99	45.77	41.63	59.19
ACF·RankAvg	18.54	43.04	38.81	55.81
ACF-ts·SimAvg	16.59	39.65	37.78	54.00
ACF-ts·GeoAvg	19.43	38.21	38.95	56.67
ACF-ts·KmeansAvg	15.11	43.79	39.51	59.47
ACF-ts·MSSEAvg	19.01	45.41	40.98	62.17
ACF-ts·RankAvg	15.04	40.84	37.34	54.73
		Male population		
Lee-Carter	28.28	40.85	49.07	65.29
ACF·AIO	28.28	39.54	46.90	65.76
ACF·GeoInfo	23.75	37.43	41.41	64.02
ACF·kmeans	24.35	36.70	42.31	58.82
ACF·RankAvg	21.49	37.15	37.13	54.52
ACF-ts·SimAvg	23.00	33.67	36.66	56.69
ACF-ts·GeoAvg	26.74	35.76	37.97	57.63
ACF-ts·KmeansAvg	24.80	35.82	39.69	58.38
ACF-ts·MSSEAvg	24.08	39.64	40.50	56.40
ACF-ts·RankAvg	24.40	31.61	35.58	56.11

the resulting sequences of test SSEs, $\{e(t), t = 2003, ..., 2010\}$ from both models. If Model A is concluded to be significantly worse than Model B by the DM test with a p value smaller than .05, we count it as a win of Model B over Model A in predicting the mortality of the target population. If Model B is confirmed to be significantly worse than Model A by the DM test, we count it as a win of Model A over Model B. There are scenarios in which both tests lead to p values larger than .05 and neither of the two models is considered to win the other in statistical significance.

Table 3 reports the number of wins for one model (from the group of **BMBE** methods) over another (from the group of benchmark models or the ACF.RankAvg model). Each cell of the table contains two integers recording the comparative results of the model in the row versus the one in the column. The first integer is the number of wins by the model in the row over the one in the column. The second integer is the number of wins by the model in the column. For example, "(23, 1)" in the first column in the female population panel means that the ACF-ts·SimAvg model wins 23 times over the Lee-Carter model and the Lee-Carter model wins over the ACF-ts·SimAvg model only once among all of the 24 comparisons for female populations. As shown in Table 3, the BMBE-based predictive models, except ACF-ts·MSSEAvg, perform significantly better than the benchmark models in terms of the number of wins. The superiority of ACF-ts·SimAvg and ACF-ts·RankAvg is evident when compared with the four benchmark models because these two models frequently win among all 24 comparisons for both female and male populations. Furthermore, the results for both genders indicate that ACF-ts·RankAvg wins ACF·RankAvg more frequently, which further reinforces the benefits of including the time shift component (i.e., the parameter Δt) in the ACF base learner.

4.3. Empirical Study with the CBD-ts Model

In this empirical study, we investigate the forecasting performance of the BMBE method for seniors with ages ranging from 55 to 90. The CBD-type models are known to provide an improvement to the Lee-Carter/APC-type models for modeling and predicting the mortality of a senior group because of the relatively simple log-linear structure of the mortality curve and parsimonious age effects; see Cairns et al. (2009, 2011). We use the CBD model and its extension with the time shift component as the base learner in our BMBE method and compare the resulting forecasting performance with the classic CBD model (Cairns, Blake, and Dowd 2006), which is also known as model M5 in Cairns et al. (2009).

4.3.1. Prediction Models

We consider all of the averaging strategies except MSSEAvg as described in Section 3 and use them with CBD models with the time shift component. We do not consider MSSEAvg in this empirical study because it performs quite poorly as confirmed in the previous empirical study with the ACF-type base learner. We label the resulting predictive models CBD-ts·SimAvg, CBD-ts·GeoAvg, CBD-ts·KmeansAvg, and CBD-ts·RankAvg, respectively. To study the value of adding a time shift component in the base learner, we also consider the CBD·RankAvg method, which is a special case of a CBD-

TABLE 3
Number of Wins for Comparison between the BMBE Approaches and the Benchmark ACF Models from DM Tests

	Lee-Carter	ACF·AIO	ACF·GeoInfo	ACF·kmeans	ACF·RankAvg
			Female population		
ACF-ts·SimAvg	(23, 1)	(22, 1)	(17, 0)	(16, 3)	(9, 3)
ACF-ts·GeoAvg	(13, 0)	(15, 1)	(13, 3)	(12, 4)	(3, 9)
ACF-ts·KmeansAvg	(15, 1)	(15, 2)	(15, 3)	(11, 3)	(5, 8)
ACF-ts·MSSEAvg	(10, 4)	(10, 4)	(9, 5)	(6, 8)	(2, 12)
ACF-ts·RankAvg	(21, 2)	(19, 2)	(16, 1)	(14, 2)	(10, 4)
			Male population		
ACF-ts·SimAvg	(24, 0)	(20, 0)	(16, 4)	(15, 4)	(8, 9)
ACF-ts·GeoAvg	(16, 3)	(13, 3)	(10, 9)	(11, 7)	(5, 12)
ACF-ts·KmeansAvg	(19, 0)	(17, 0)	(10, 5)	(10, 9)	(6, 12)
ACF-ts·MSSEAvg	(13, 4)	(10, 5)	(7, 8)	(8, 9)	(3, 14)
ACF-ts·RankAvg	(24, 0)	(21, 0)	(14, 3)	(17, 3)	(11, 5)

Note: In each cell, the first integer indicates the number of wins by the model in the row over the model in the column out of 24 comparisons and the second integer is the number of wins by the model in the column over the one in the row.

ts·RankAvg method with Δt fixed at zero in the base learner. For comparison, we consider the classic CBD model as a benchmark.

4.3.2. Prediction Performance Evaluation Based on Test SSEs As in the previous empirical study, we use the overall test SSE $\sum_{t=2003}^{2010} e(t)$ as a measure of prediction accuracy, where

$$e(t) = \sum_{x=55}^{90} [\text{logit } q_1(x,t) - \text{logit } \hat{q}_1(x,t)]^2, \quad t = 2003, ..., 2010;$$

$q_1(x,t)$ is the probability that an age-x individual from the target population dies between year t and $t+1$ given the individual is alive at time t, and $\hat{q}_1(x,t)$ is its prediction.

Table 4 summarizes the prediction performance of the various models in the second empirical study. The CBD-ts·RankAvg method is a clear winner over the classic CBD model for both female and male populations. The performance of the SimAvg strategy is disappointing and worse than that of the classic CBD model, which is in contrast to its outstanding performance in the first empirical study. The success of the SimAvg strategy in the first empirical analysis is due to the fact that the trajectories of age-aggregated logarithmic morality rate share similar negative slopes as shown in Figure 1 and a ACF-ts model works as a reasonably well-specified base learner for most pairs of populations. The parallel trajectories of the aggregate logarithmic mortality rates within the 24 populations were also confirmed by the MDS-based clustering analysis described in Section 4.1. However, the current empirical study considers the trajectories of age-aggregated logit$q_j(x,t)$ over ages 50 to 90, and these trajectories are not parallel to each other to the same degree as in the first empirical study. This implies that the CBD-ts model could potentially be a misspecified base learner for a population pair, and the biases of forecast arising in each misspecified base learner accumulate and lead to deteriorated prediction results in the SimAvg strategy. The misspecification issue is much less severe for the GeoAvg and KmeansAvg strategies because they only average over a small subset of populations.

The superiority of the RankAvg strategy is apparent. The test SSEs from this strategy (from both CBD-ts.RankAvg and CBD.RankAvg) are smaller than those from the CBD model for both male and female populations. This strategy performs well in both empirical studies and exhibits some robustness in its outperformance. It does not require any pregrouping and uses a fully

TABLE 4
Summary Statistics of Test SSE's Comparing the Prediction Performance of BMBE-Based Approaches using CBD or CBD-ts as the Base Learner versus the CBD Model

	First quartile	Median	Mean	Third quartile
		Female population		
CBD	3.61	5.87	7.37	11.67
CBD·RankAvg	3.76	6.06	6.31	7.66
CBD-ts·SimAvg	5.13	7.96	10.28	13.32
CBD-ts·GeoAvg	4.60	6.50	8.76	11.95
CBD-ts·KmeansAvg	3.97	5.76	7.61	11.70
CBD-ts·RankAvg	3.18	5.76	6.94	10.32
M6	5.37	7.53	8.45	10.40
M6-ts·RankAvg	8.14	33.96	59.19	103.29
		Male population		
CBD	2.32	2.88	3.08	3.63
CBD·RankAvg	2.08	2.93	2.94	3.44
CBD-ts·SimAvg	2.08	3.04	5.95	4.38
CBD-ts·GeoAvg	2.12	2.60	2.84	3.69
CBD-ts·KmeansAvg	2.22	2.54	2.84	3.56
CBD-ts·RankAvg	2.04	2.67	2.85	3.38
M6	2.13	3.14	3.20	3.73
M6-ts·RankAvg	3.70	7.49	16.64	14.97

data-driven mechanism to identify a subset of auxiliary populations for the final forest. When an auxiliary population demonstrates a different development pattern from the target population, the prediction based on the corresponding CBD-ts model may lead to a large prediction error and the RankAvg strategy tends to rank this auxiliary population low and eventually screen it out in the cross-validation procedure.

Furthermore, a comparison between CBD·RankAvg and CBD-ts·RankAvg shows that the inclusion of the time shift component in the base learner improves the results slightly for male populations but not for female populations. The results of the two empirical studies suggest that there is a trade-off between prediction accuracy and robustness. When we include a time shift term in the base learner, we implicitly assume that the trajectory of one population is roughly a shift of the other. If the assumption is satisfied, the base learner with the time shift term is more efficient and the final forecast is more accurate and stable. Nevertheless, if the assumption is severely violated, we pay a heavy price in the form of a biased base learner and get a deteriorated final forecast. The base learner without time shift terms gives mediocre prediction results if the time shift assumption is satisfied, and they are not subjected to being overly sensitive to misspecification.

Evaluation Based on One-Sided DM Tests We also apply the DM test to compare the performance of the various predictive models and report the results in Table 5. The table suggests that all of the BMBE methods still beat the CBD model in terms of the number of wins, except when CBD-ts·GeoAvg is applied to female populations. The DM test also confirms the superiority of CBD-ts·RankAvg among all predictive models.

4.4. Inclusion of Cohort Effect

The cohort effect, which addresses the mortality differences between people with different years of birth, has been widely documented in the mortality literature; see Renshaw and Haberman (2006), Cairns et al. (2009), Plat (2009), and Hunt and Blake (2021). In this section, we study the possible influence of the cohort effect on mortality forecasting under our BMBE framework.

4.4.1. Inclusion of Cohort Effect in ACF-Type Model

We consider the following ACF-type model with a cohort effect as the base learner:

$$\log m_1(x,t) = a_1(x) + B(x)K(t) + b_1(x)k_1(t) + \gamma(x,t) + \epsilon_1(x,t), \tag{6}$$

$$\log m_2(x,t) = a_2(x) + B(x)K(t) + b_2(x)k_2(t) + \gamma(x,t) + \epsilon_2(x,t). \tag{7}$$

This specification can be viewed as an extension to the ACF-ts model in (1) and (2) with the inclusion of the extra cohort effect terms $\gamma(x,t)$ and a further constraint that $\Delta t = 0$. The model in (6) and (7) can also be viewed as a specific form of the cohort extension of the Poisson common factor model in Yang, Li, and Balasooriya (2016). We follow their calibration and forecasting procedure and compare its predictive performance with the APC model, a commonly used substructure of the Renshaw and Haberman model (Renshaw and Haberman 2006), corresponding to $\beta_x^{(0)} = 1$ and $\beta_x^{(1)} = 1$.

Table 6 summarizes the test SSEs from the single-population APC model and the BMBE method by using the model in (6) and (7) as the base learner (labeled as ACF-cohort·RankAvg because the RankAvg averaging strategy is applied). A

TABLE 5
Number of Wins for Comparisons between a CBD-ts Model and the CBD Model Based on a Pair of One-Sided DM Tests

	CBD	
	Female	Male
CBD·RankAvg	(12, 9)	(9, 8)
CBD-ts·SimAvg	(10, 7)	(12, 9)
CBD-ts·GeoAvg	(7, 8)	(12, 6)
CBD-ts·KmeansAvg	(11, 5)	(6, 5)
CBD-ts·RankAvg	(11, 3)	(10, 5)

Note: In each cell, the first integer indicates the number of wins of the model in the row over the model in the column out of 24 comparisons and the second integer is the number of wins of the model in the column over the one in the row.

comparison across different models in Tables 2 and 6 indicates that adding the cohort effect term to the base learner does not lead to a clear improvement in test SSE. Indeed, the average test SSE over all of the target populations becomes larger. The BMBE method, however, continues to show an obvious outperformance over the APC model.

4.4.2. Inclusion of Cohort Effect in CBD-Type Model

We consider the following CBD-type model with a cohort effect as the base learner:

$$\log\left[\frac{q_j(x,t)}{1-q_j(x,t)}\right] = K(t) + (x-\bar{x})k_1(t) + \gamma_1(x,t) + \epsilon_1(x,t), \tag{8}$$

$$\log\left[\frac{q_j(x,t)}{1-q_j(x,t)}\right] = K(t-\Delta t) + (x-\bar{x})k_2(t) + \gamma_2(x,t) + \epsilon_2(x,t), \tag{9}$$

where $q_j(x,t)$, $K(t)$, $k_j(t)$, $\bar{x}$, $\epsilon_j(x,t)$, and Δt carry the same meaning as the specification in (3) and (4) for CBD-ts model. This specification extends the original single-population M6 model in Cairns et al. (2009) to a bivariate-population model by introducing a common $K(t)$ and a time shift term to facilitate borrowing information from the auxiliary population across time. We label this model M6-ts. The difference between M6-ts and the CBD-ts model is the inclusion of the extra cohort effect terms $\gamma_j(x,t)$, $j=1, 2$.

Table 7 summarizes the test SSEs from the single-population M6 model and the BMBE method by using the model in (8) and (9) as the base learner. The RankAvg averaging strategy is applied in the BMBE method and therefore the resulting predictive model is labeled as M6-ts·RankAvg. A comparison between Tables 4 and 7 indicates that the test SSEs of M6-ts·RankAvg are significantly larger than those of the other models, especially for female populations. The inclusion of both the cohort effect term and the time shift term in the base learner is detrimental to the overall forecasting accuracy in the study.

There are considerable differences in the strength of the cohort effect across different populations, and strong cohort patterns have only been found in some populations; see H. Li, O'Hare, and Vahid (2016). Figure 2 shows two illustrative

TABLE 6
Summary Statistics of Test SSE's for BMBE Approach (Using RankAvg Averaging Strategy and ACF-Cohort Model as the Base Learner) versus the Plain APC Mode

	First quartile	Median	Mean	Third quartile
		Female population		
APC	32.55	46.34	49.44	72.55
ACF-cohort·RankAvg	15.95	41.96	39.06	55.70
		Male population		
APC	27.78	46.53	44.68	59.95
ACF-cohort·RankAvg	25.63	33.47	39.96	58.39

TABLE 7
Summary Statistics of Test SSE's for BMBE Approach (Using RankAvg Averaging Strategy and M6 as the Base Learner) versus the Plain M6 Model

	First quartile	Median	Mean	Third quartile
		Female population		
M6	5.37	7.53	8.45	10.40
M6-ts·RankAvg	8.14	33.96	59.19	103.29
		Male population		
M6	2.13	3.14	3.20	3.73
M6-ts·RankAvg	3.70	7.49	16.64	14.97

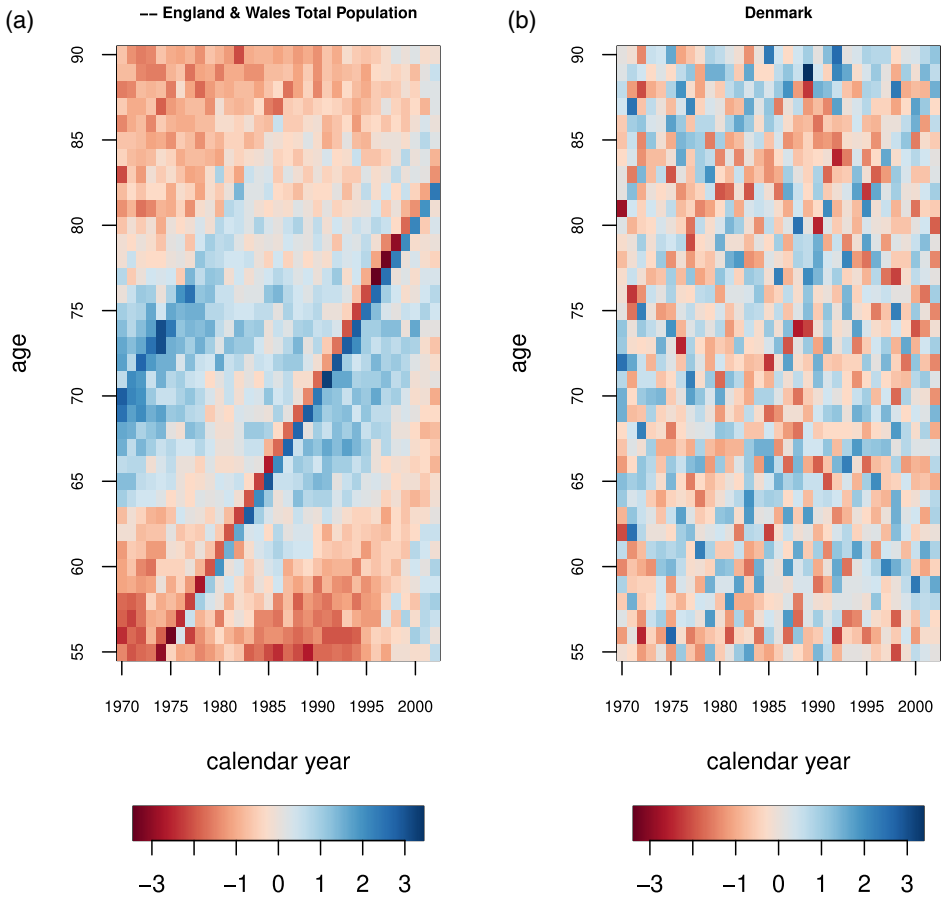


FIGURE 2. Deviance Residual of the CBD Model on Different Populations. (a) Males Aged 55 to 90 for England and Wales and (b) Males Aged 55 to 90 for Demark.

examples on the disparate strength of the cohort effect. Strong diagonal signals in the residuals of the CBD model indicate the existence of a cohort effect for the male seniors from England and Wales in Figure 2(a). In contrast, the residuals of the CBD model exhibit pure randomness, so male seniors from Denmark in Figure 2(b) do not suggest the existence of a cohort effect. Therefore, the M6-ts model containing cohort effect terms for both populations is a misspecified model for this pair of populations. Adding cohort effects to both populations in the base learner may only benefit a target population having a strong cohort effect itself and can borrow information from the ones that also have strong cohort effects in the BMBE method. It compromises the overall forecasting performance because it is a misspecified base learner as long as one in each population pair does not include a noticeable cohort effect. This suggests that a complicated parametric model with many model assumptions is not suitable to serve as a base learner under the BMBE framework.

4.5. Synthetic Study

In this section, we construct a synthetic mortality data generating process to illustrate the outperformance of our BMBE method when there is greater risk of model misspecification for a higher dimensional multiple-population model than for bivariate models.

4.5.1. Simulation Setup

We consider 24 populations with their logarithmic mortality rates described by the following model:

$$\log m_j(x,t) = a_j(x) + F_j(x,t) + f_j(x,t), \tag{10}$$

for $j = 1, 2, ..., 24$. The terms $F_j(x,t)$, although population specific, are driven by eight independent common factor

processes $C_l(x,t)$, $l = 1,2,...,8$, whereas $f_j(x,t)$ are purely population specific and independent of all eight common factor processes. For reference, we call the above model a "synthetical model" hereafter. The specific values of $a_j(x)$ and models of $f_j(x,t)$ and the eight common factor processes come from a calibration procedure with the mortality data (for years 1970–2010) of the 24 populations listed in Table 1. For each geographic group l (respectively labeled groups $1,2,...,8$ according to the order of these eight groups in Table 1) as indicated in Table 1, we fit their mortality data with an ACF model by using the regular SVD calibration procedure:

$$\log m_j^{(l)}(x,t) = a_j^{(l)}(x) + B^{(l)}(x)K^{(l)}(t) + b_j^{(l)}(x)k_j^{(l)}(t) + \epsilon_j^{(l)}(x,t), \tag{11}$$

where $K^{(l)}(t)$ is fitted with an ARIMA model, and $k_j^{(l)}$ is fitted with either an AR(1) or AR(2) model depending on which leads to a better Bayesian information criterion value. In the above, $a_j^{(l)}(x)$ is the average level of the logarithmic mortality rates for age x of population j in group l, and $\epsilon_j^{(l)}(x,t)$ are residual terms.

For each of the 24 populations in the synthetic model (10), $a_j(x)$ is taken as the corresponding $a_j^{(l)}(x)$ and $f_j(x,t)$ is assigned with the AR(1) or AR(2) model from the above ACF model calibration for the corresponding population. Furthermore, the term $F_j(x,t)$ is defined by

$$F_j(x,t) = \mathbb{L}_j'\mathbb{C}(x,t), \tag{12}$$

where $\mathbb{C}(x,t) := (C_1(x,t),...,C_8(x,t)'$ with each common factor process taken as

$$C_l(x,t) = B^{(l)}(x)K^{(l)}(t), \quad l = 1,2,...,8,$$

and the $\mathbb{L}_j'$ is the jth row from the following loading matrix in three blocks:

$$\mathbb{L} = \begin{pmatrix} M_1 \\ M_2 \\ M_3 \end{pmatrix}$$

with

$$M_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.9 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.8 & 0.1 & 0.1 & 0 & 0 & 0 & 0 & 0 \\ 0.7 & 0.1 & 0.1 & 0.1 & 0 & 0 & 0 & 0 \\ 0.6 & 0.1 & 0.1 & 0.1 & 0.1 & 0 & 0 & 0 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0 & 0 \\ 0.4 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0 \\ 0.3 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0.9 \\ 0 & 0 & 0 & 0 & 0 & 0.1 & 0.1 & 0.8 \\ 0 & 0 & 0 & 0 & 0.1 & 0.1 & 0.1 & 0.7 \\ 0 & 0 & 0 & 0.1 & 0.1 & 0.1 & 0.1 & 0.6 \\ 0 & 0 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.4 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.3 \end{pmatrix}$$

and

$$M_3 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.9 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0.1 & 0.8 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0.1 & 0.7 & 0.1 & 0.1 & 0 & 0 \\ 0 & 0.1 & 0.1 & 0.6 & 0.1 & 0.1 & 0 & 0 \\ 0 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0 \\ 0.1 & 0.1 & 0.1 & 0.4 & 0.1 & 0.1 & 0.1 & 0 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.1 & 0.1 & 0.1 & 0.1 \end{pmatrix}.$$

Note that each row of the loading matrix $\mathbb{L}$ sums to one, and it is designed such that $F_j(x,t)$s share greater similarity for two neighboring populations within each of the three blocks and less similarity when they are further apart from each other. So, a bivariate ACF model for two neighboring populations from the same block has less misspecification compared with an ACF model modeling for more than two populations.

4.5.2. Simulation Results

We simulate mortality data $\log m_j(x,t)$ from the synthetic model (10) as described in the preceding section 500 times independently. Each simulated dataset contains mortality data for the 24 populations of 40 years. We split the data into a training set with a length of 32 years and a test set with a length of 8 years. We apply our **ACF·RankAvg** method with the training dataset to develop a mortality prediction model and compute the test SSE for each of the 24 populations based on the 8-year test dataset. In addition, we apply the following benchmark models for comparative analysis:

- **Lee-Carter**: For each simulated dataset, the Lee-Carter model is fitted to the 24 populations separately.
- **ACF·AIO**: For each simulated dataset, the ACF model is fitted to the 24 populations jointly.
- **ACF·kmeans**: For each simulated dataset, a k-means clustering is first utilized to determine the cluster members and the ACF model is fitted to each cluster. We use the recommended value of k from the `NbClust` function in `R` package `NbClust` (Charrad et al. 2014) to determine the number of clusters, following Tsai and Cheng (2021).

To demonstrate the relative performance of these predictive models, we report two sets of statistics from the simulation results for the test SSEs. In the first set of statistics, we compute the average test SSE from the 500 independent simulations for each of the 24 populations. This gives us 24 average SSE for the 24 populations. We then compute their first quartile, median, mean, and third quartile and present them in the top panel of Table 8 for each predictive model. In the second set of statistics, we sum the SSEs for all 24 populations in each simulation. This gives us 500 overall SSEs. We also compute their first quartile, median, mean, and third quartile and display their values in the bottom panel of Table 8.

Both pannels in Table 8 confirm the strong outperformance of the BMBE method (ACF.RankAvg) relative to the other predictdive methods because it has the smallest SSEs in general. The Lee-Carter model has the worst test SSEs because it does not borrow any information from other populations in predicting the mortality of a target population. Both the ACF·AIO and

TABLE 8
Summary Statistics of Test SSEs Comparing the Prediction Performance of the BMBE Method versus Benchmark Models

	First quartile	Median	Mean	Third quartile
		Over 24 populations		
Lee-Carter	2.98	6.50	8.12	10.95
ACF·AIO	3.92	6.35	7.67	9.27
ACF·kmeans	4.33	6.07	7.88	9.24
ACF·RankAvg	3.08	4.52	5.48	6.72
		Over 500 simulations		
Lee-Carter	181.4	205.4	194.9	205.4
ACF·AIO	160.8	167.2	184.1	208.3
ACF·kmeans	166.9	192.2	189.1	211.2
ACF·RankAvg	125.2	135.2	131.5	135.2

the ACF·kmeans methods borrow information across populations through the joint modeling, but they have a more severe misspecification issue than the BMBE method. Accordingly, their SSEs are generally in between those for the Lee-Carter model and those for the BMBE-based model.

5. CONCLUDING REMARKS

We propose a general and flexible ensemble framework for mortality forecasting that allows for borrowing information across populations via a cascade of bivariate mortality models, which we call the BMBE method. The framework ensembles forecasts from all of the bivariate mortality models by using various averaging strategies to generate the final forecast, among which the rank and average averaging strategy yields robust outperformance in forecast accuracy. The "Rank and Average" strategy also turns out to outperform benchmark models in our empirical studies. Salient advantages of the rank and average strategy lie in its having no requirement of pregrouping populations and its ability to recognize useful information and exclude irrelevant noise in a data-driven manner. We also study the effect of a time shift component and a cohort effect term on the base learner on the resulting mortality forecasting accuracy within the ensemble framework. Our empirical studies suggest that avoiding severe misspecification in the base learner is the key to good performance when utilizing the BMBE approach.

There are several aspects of this study that are worth exploration in further improving the model averaging framework:

- First, though ACF and CBD models are used as the base learners in the empirical studies in this article, other bivariate-population models are worth consideration as well to be used in the model averaging framework; for example, the vector autoregression and vector error correction models introduced by Zhou et al. (2014).
- Second, because the model averaging procedure is computationally efficient, it can be extended to more complex scenarios. For instance, when interest lies in forecasting mortality rates of a particular age group (i.e., infant, teenage, or senior), a large number of candidate populations of different regions, genders, and ages form the pool of auxiliary populations.
- Third, although the general BMBE framework does not impose any constraints on the base learner, a base learner that captures more desirable characteristics of the mortality data is likely to be more conducive to improving the prediction accuracy of the general framework. It could be a rewarding exercise to consider a set of base learner candidates (instead of a fixed base learner for every population pair as we do in the present article) and apply a data-driven model selection procedure for the fit of each population pair before implementing a model averaging strategy for the final forecast. An automated procedure of identifying a suitable base learner for each population pair is imperative if a large set of base learner candidates will be implemented, particularly when the pool of auxiliary populations is also large.

For the design of base learner candidates, other model features, in addition to time shifting and cohort effect, can also be considered. For instance, it would be interesting to add a scale factor or exposure adjustment factor to the common trend term in the AFT-ts and CBD-ts models to accommodate the scenarios when the aggregate mortality trajectories of two populations have different slopes. In addition, we may create base learner candidates by extending the myriad single-population models in Cairns et al. (2009) to a bivariate-population version. It would also be worth exploring possible choices of the functional form for the common mortality trend term in the ACF model other than an age–time separation form; that is, $B(x)K(t)$. Furthermore, because the strength of the cohort effect varies from population to population, it could also be interesting to consider a base learner candidate in the set of candidate pool with an "asymmetric" structure where a cohort effect term is included for one population but not for the other.

ACKNOWLEDGMENTS

The authors are grateful for the time spent by the editor in handling our article. The authors are also thankful for the valuable comments from three anonymous reviewers that have led to substantial improvement in the contents and presentation of the article.

FUNDING

The authors acknowledge funding support from the Canadian Institute of Actuaries (No. CS000215).

ORCID

Liqun Diao http://orcid.org/0000-0003-2965-8220
Yechao Meng http://orcid.org/0000-0002-3458-1021
Chengguo Weng http://orcid.org/0000-0002-0862-7743
Tony Wirjanto http://orcid.org/0000-0003-1324-9131

REFERENCES

Brainerd, E., and D. M Cutler. 2005. Autopsy on an empire: Understanding mortality in Russia and the former Soviet Union. *Journal of Economic Perspectives* 19 (1):107–30.

Bühlmann, P., and B. Yu. 2002. Analyzing bagging. *Annals of Statistics* 30 (4):927–61.

Cairns, A. J. G., D. Blake, and K. Dowd. 2006. A two-factor model for stochastic mortality with parameter uncertainty: Theory and calibration. *Journal of Risk and Insurance* 73 (4):687–718. doi:10.1111/j.1539-6975.2006.00195.x

Cairns, A. J. G., D. Blake, K. Dowd, G. D. Coughlan, D. Epstein, and M. Khalaf-Allah. 2011. Mortality density forecasts: An analysis of six stochastic mortality models. *Insurance: Mathematics and Economics* 48 (3):355–67. doi:10.1016/j.insmatheco.2010.12.005

Cairns, A. J. G., D. Blake, K. Dowd, G. D. Coughlan, D. Epstein, A. Ong, and I. Balevich. 2009. A quantitative comparison of stochastic mortality models using data from England and Wales and the United States. *North American Actuarial Journal* 13 (1):1–35. doi:10.1080/10920277.2009.10597538

Chang, L., and Y. Shi. 2022. Forecasting mortality rates with a coherent ensemble averaging approach. *ASTIN Bulletin* 2022:1–27. doi:10.1017/asb.2022.23

Charrad, M., N. Ghazzali, V. Boiteau, and A. Niknafs. 2014. NbClust: an R package for determining the relevant number of clusters in a data set. *Journal of statistical software* 61:1–36.

Claeskens, G., and N. L. Hjort. 2008. *Model selection and model averaging*. Cambridge: Cambridge University Press.

Cox, M. A. A., and T. F. Cox. 2008. Multidimensional scaling. In *Handbook of data visualization*, eds. C.-H. Chen, W. Hardle and A. Unwin, 315–47. Berlin: Springer.

DeMiguel, V., L. Garlappi, and R. Uppal. 2009. Optimal versus naive diversification: How inefficient is the 1/*n* portfolio strategy? *Review of Financial Studies* 22 (5):1915–953. doi:10.1093/rfs/hhm075

Diao, L., Y. Meng, and C. Weng. 2021. A DSA algorithm for mortality forecasting. *North American Actuarial Journal* 25 (3):438–58. doi:10.1080/10920277.2020.1806884

Diebold, F. X., and R. S. Mariano. 2002. Comparing predictive accuracy. *Journal of Business and Economic Statistics* 20 (1):134–44. doi:10.1198/073500102753410444

Dowd, K., A. J. G. Cairns, D. Blake, G. D. Coughlan, and M. Khalaf-Allah. 2011. A gravity model of mortality rates for two related populations. *North American Actuarial Journal* 15 (2):334–56. doi:10.1080/10920277.2011.10597624

Hansen, B. E., and J. S. Racine. 2012. Jackknife model averaging. *Journal of Econometrics* 167 (1):38–46. doi:10.1016/j.jeconom.2011.06.019

Harvey, D., S. Leybourne, and P. Newbold. 1997. Testing the equality of prediction mean squared errors. *International Journal of Forecasting* 13 (2):281–91. doi:10.1016/S0169-2070(96)00719-4

Hatzopoulos, P., and S. Haberman. 2013. Common mortality modeling and coherent forecasts. An empirical analysis of worldwide mortality data. *Insurance: Mathematics and Economics* 52 (2):320–37. doi:10.1016/j.insmatheco.2012.12.009

Hunt, A., and D. Blake. 2021. On the structure and classification of mortality models. *North American Actuarial Journal* 25 (suppl. 1):S215–S34. doi:10.1080/10920277.2019.1649156

Hyndman, R. J., and Y. Khandakar. 2007. Automatic time series for forecasting: the forecast package for R (No. 6/07). Monash University, Department of Econometrics and Business Statistics.

Hyndman, R. J., H. Booth, and F. Yasmeen. 2013. Coherent mortality forecasting: The product-ratio method with functional time series models. *Demography* 50 (1):261–83. doi:10.1007/s13524-012-0145-5

Janssen, F., L. J. G. van Wissen, and A. E. Kunst. 2013. Including the smoking epidemic in internationally coherent mortality projections. *Demography* 50 (4):1341–362. doi:10.1007/s13524-012-0185-x

Kessy, S. R., M. Sherris, A. M. Villegas, and J. Ziveyi. 2022. Mortality forecasting using stacked regression ensembles. *Scandinavian Actuarial Journal* 2022 (7):591–626.

Kleinow, T. 2015. A common age effect model for the mortality of multiple populations. *Insurance: Mathematics and Economics* 63:147–52. doi:10.1016/j.insmatheco.2015.03.023

Lee, R. D., and L. R. Carter. 1992. Modeling and forecasting U.S. mortality. *Journal of the American Statistical Association* 87 (419):659–71. doi:10.1080/01621459.1992.10475265

Li, H., C. O'Hare, and F. Vahid. 2016. Two-dimensional kernel smoothing of mortality surface: An evaluation of cohort strength. *Journal of Forecasting* 35 (6):553–63. doi:10.1002/for.2399

Li, J. 2013. A Poisson common factor model for projecting mortality and life expectancy jointly for females and males. *Population Studies* 67 (1):111–26. doi:10.1080/00324728.2012.689316

Li, J. S.-H., W.-S. Chan, and R. Zhou. 2017. Semicoherent multipopulation mortality modeling: The impact on longevity risk securitization. *Journal of Risk and Insurance* 84 (3):1025–65. doi:10.1111/jori.12135

Li, J. S.-H., and M. R. Hardy. 2011. Measuring basis risk in longevity hedges. *North American Actuarial Journal* 15 (2):177–200. doi:10.1080/10920277.2011.10597616

Li, J. S.-H., R. Zhou, and M. Hardy. 2015. A step-by-step guide to building two-population stochastic mortality models. *Insurance: Mathematics and Economics* 63:121–34. doi:10.1016/j.insmatheco.2015.03.021

Li, N., and R. Lee. 2005. Coherent mortality forecasts for a group of populations: An extension of the Lee-Carter method. *Demography* 42 (3):575–94. doi:10.1353/dem.2005.0021

Milidonis, A., and M. Efthymiou. 2017. Mortality leads and lags. *Journal of Risk and Insurance* 84 (suppl. 1):495–514. doi:10.1111/jori.12187
Plat, R. 2009. On stochastic mortality modeling. *Insurance: Mathematics and Economics* 45 (3):393–404.
Raftery, A. E., D. Madigan, and J. A. Hoeting. 1997. Bayesian model averaging for linear regression models. *Journal of the American Statistical Association* 92 (437): 179–91. doi:10.1080/01621459.1997.10473615
Renshaw, A. E., and S. Haberman. 2006. A cohort-based extension to the Lee-Carter model for mortality reduction factors. *Insurance: Mathematics and Economics* 38 (3):556–70. doi:10.1016/j.insmatheco.2005.12.001
Richman, R., and M. V. Wüthrich. 2021. A neural network extension of the Lee-Carter model to multiple populations. *Annals of Actuarial Science* 15 (2): 346–66. doi:10.1017/S1748499519000071
Russolillo, M., G. Giordano, and S. Haberman. 2011. Extending the Lee-Carter model: A three-way decomposition. *Scandinavian Actuarial Journal* 2011 (2):96–117. doi:10.1080/03461231003611933
Shair, S., S. Purcal, and N. Parr. 2017. Evaluating extensions to coherent mortality forecasting models. *Risks* 5 (1):16. 10.3390/risks5010016
Shang, H. L. 2012. Point and interval forecasts of age-specific life expectancies: A model averaging approach. *Demographic Research* 27:593–644. doi:10.4054/DemRes.2012.27.21
Shang, H. L. 2016. Mortality and life expectancy forecasting for a group of populations in developed countries: A multilevel functional data method. *Annals of Applied Statistics* 10 (3):1639–672.
Shang, H. L., and H. Booth. 2020. Synergy in fertility forecasting: Improving forecast accuracy through model averaging. *Genus* 76 (1):1–23. doi:10.1186/s41118-020-00099-y
Shang, H. L., and S. Haberman. 2018. Model confidence sets and forecast combination: An application to age-specific mortality. *Genus* 74 (1):1–23. doi:10.1186/s41118-018-0043-9
Shang, H. L., and R. J. Hyndman. 2017. Grouped functional time series forecasting: An application to age-specific mortality rates. *Journal of Computational and Graphical Statistics* 26 (2):330–43. doi:10.1080/10618600.2016.1237877
Tsai, C. C.-L., and E. S. Cheng. 2021. Incorporating statistical clustering methods into mortality models to improve forecasting performances. *Insurance: Mathematics and Economics* 99:42–62. doi:10.1016/j.insmatheco.2021.03.005
Wang, C.-W., and S. S. Yang. 2013. Pricing survivor derivatives with cohort mortality dependence under the Lee-Carter framework. *Journal of Risk and Insurance* 80 (4):1027–56. doi:10.1111/j.1539-6975.2012.01488.x
Yang, B., J. Li, and U. Balasooriya. 2016. Cohort extensions of the Poisson common factor model for modelling both genders jointly. *Scandinavian Actuarial Journal* 2016 (2):93–112. doi:10.1080/03461238.2014.908411
Zhou, R., Y. Wang, K. Kaufhold, J. S.-H. Li, and K. S. Tan. 2014. Modeling period effects in multi-population mortality models: Applications to Solvency II. *North American Actuarial Journal* 18 (1):150–67. doi:10.1080/10920277.2013.872553

Discussions on this article can be submitted until January 1, 2024. The authors reserve the right to reply to any discussion. Please see the Instructions for Authors found online at http://www.tandfonline.com/uaaj for submission instructions.

APPENDIX A. CALIBRATION AND EXTRAPOLATION OF BASE LEARNERS

A.1. Calibration of the ACF-ts Model

Following are the specific steps for the calibration and extrapolation of the ACF-ts model with Δt being fixed:

1. We shift data of the auxiliary population (i.e., population 2) Δt years backward along the timeline if $\Delta t \geq 0$ or Δt years forward if $\Delta t < 0$. We then aggregate the data by using age–time-specific population size as the weighting variable. Specifically, if we let $E_j(x,t)$ denote the exposure number for age x and time t from population j, $j = 1, 2$, then the aggregate death rates are computed as

$$\frac{E_1(x,t)m_1(x,t) + E_2(x,t)m_2(x,t)}{E_1(x,t) + E_2(x,t)}.$$

The resulting aggregated data span over the time horizon $[-\Delta t,\ T]$ if $\Delta t \geq 0$ or $[0,\ T - \Delta t]$ if $\Delta t < 0$; that is,

- when $\Delta t \geq 0$:
  - over $[-\Delta t,\ 0)$, it only contains data from the auxiliary $(j = 2)$;
  - over $[0,\ T - \Delta t]$, it contains the weighted average of the data from both the target $(j = 1)$ and the auxiliary $(j = 2)$;
  - over $(T - \Delta t,\ T]$, it only contains data from the target $(j = 1)$.
- when $\Delta t < 0$:
  - over $[0,\ -\Delta t)$, it only contains data from the target $(j = 1)$;
  - over $[-\Delta t,\ T]$, it contains the weighted average of the data from both the target $(j = 1)$ and the auxiliary $(j = 2)$;
  - over $(T,\ T - \Delta t]$, it only contains data from the auxiliary $(j = 2)$.

2. Based on the aggregated data from Step 1, we calibrate $B(x)$ and $K(t)$ as the first left and right singular vectors from an SVD procedure. The length of calibrated sequence $K(t)$ depends on the value of Δt. The time index of realized $K(t)$ is $\{-\Delta t, -\Delta t+1, ..., T\}$ for $\Delta t \geq 0$ and $\{0, 1, ..., T-\Delta t\}$ for $\Delta t < 0$.
3. Denote the residuals as $R_1(x,t) = [\log m_1(x,t) - B(x)K(t)]$ and $R_2(x,t) = [\log m_2(x,t) - B(x)K(t-\Delta t)]$. Then, we calibrate $a_j(x)$ as an average of the residual rates over the modeling period for each population and age:
$$a_j(x) = \frac{1}{T+1}\sum_{t=0}^{T} R_j(x,t), \quad j = 1, 2, \text{ and } \quad x = x_1, ..., x_n.$$
4. For each $j = 1, 2$, apply an SVD procedure further to $R_j(x,t) - a_j(x)$ to get $b_j(x)$ and $k_j(t)$.
5. Fit the sequence $K(t)$ by an RWD and each sequence $k_j(t)$ with an ARIMA model using the `auto.arima` function from the `R` package `forecast`.
6. Follow the conventional extrapolation paradigm for mortality forecasting. We obtain projections of $K(t)$ and $k_1(t)$ into future years t and then use Equation (1) to obtain forecasts by setting the white noise terms to zero. The prediction of $K(t)$ is obtained either from the calibration step or by an extrapolating procedure, depending on the value of Δt. If $\Delta t \geq 0$, the aggregated data from Step 1 span over the time horizon $[-\Delta t, T]$, and in this case, the prediction of $K(t)$ for any future year $t > T$ should be obtained via extrapolating the established time series model. In contrast, if $\Delta t < 0$, the aggregated data from Step 1 span over the time horizon $[0, T-\Delta t]$, where we note $T - \Delta t > T$. In this case, we directly input the calibrated values of $K(t)$ obtained in Step 2 into Equation (1) for mortality forecasts over the period $[T, T-\Delta t]$ and apply the extrapolation procedure for prediction beyond time $T - \Delta t$.

A.2. Calibration of the CBD-ts Model

The model extrapolation of the CBD-ts model follows Steps 5 and 6 for the ACF-ts model described in Appendix A.1. Following are the specific steps in our calibration for the CBD-ts model with a fixed Δt :

1. Assume that the number of deaths at age x in year t for the jth population, denoted by $d_j(x,t)$, follows a binomial distribution with an exposure number $E_j(x,t)$ and 1-year death probability $q_j(x,t)$; that is, $d_j(x,t) \sim Bin[E_j(x,t), q_j(x,t)]$.
2. With the realized number of deaths $d_j(x,t)$ and the exposure number $E_j(x,t)$, we compute the log-likelihood function:
$$\ell = \sum_{j=1}^{2}\sum_{l=1}^{n}\sum_{t\in\mathbb{S}} \ell(d_j(x_l,t), q_j(x_l,t)), \tag{A.1}$$
where
$$\ell(d_j(x,t), q_j(x,t)) \quad \propto \quad d_j(x,t)\log\big[q_j(x,t)\big] + \big[E_j(x,t) - d_j(x,t)\big]\log\big[1 - q_j(x,t)\big].$$
3. From Equations (3) and (4), we have the following two expressions:
$$q_1(x,t) = \text{expit}[K(t) + (x-\bar{x})k_1(t)], \tag{A.2}$$
$$q_2(x,t) = \text{expit}[K(t-\Delta t) + (x-\bar{x})k_2(t)], \tag{A.3}$$
where $\text{expit}(u) = \frac{\exp(u)}{1+\exp(u)}$.
4. Calibrate the $K(t)$ and $k_j(t)$ sequences by plugging (A.2) and (A.3) into the log-likelihood (A.1) and then maximizing it with respect to $K(t)$ and $k_j(t)$ using the Newton-Raphson iterative procedure.